



Mini-Capstone: PRD Assembly

Week 3 • Day 4 • Requirements Engineering & Mini-Capstone

TPM Academy — Non-AI keystone day



Day 4 Objectives

- Write **§8 Metrics & Validation** tied to your Tier Sheet and NS
- Write **§9 Risks & Open Questions** — "no risks" is a fail
- Write **§10 Dependencies** with named owners
- Write **§11 Out-of-Scope Follow-ups** — the negotiation tool
- Run an **integration pass** — coherence + AI-prose detection — and lock the version for Friday



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Last day non-AI; Friday pairings posted
09:15 – 10:30	Block 1 + Activity 1	§8 Metrics & Validation
10:45 – 12:00	Block 2 + Activity 2	§9 Risks, Questions, Assumptions
13:00 – 14:15	Block 3 + Activity 3	§§10 + 11 Dependencies + Out-of-scope
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Integration pass + lock

Block 1 — §8 Metrics & Validation

Reconnect the PRD to the strategy



§8 Anatomy

- **Primary metric** — one number, with target + defense
- **Counter-metric** — the guardrail from your NS Defense Card
- **Secondary metrics** (3 max) — directional indicators
- **Validation plan** — pre-launch / 7 days / 30 days / 90 days
- **Definition of "good enough"** — the success bar in plain words

Coherence check: the metrics here must match Week-2's Tier Sheet. If they don't, one of them is wrong.



Worked §8 (FieldPulse)

```
## 8. Metrics & validation
```

```
<h3 id="primary-metric">Primary metric</h3>
```

```
<blockquote>
```

```
<p>Median reconcile-flow duration per dispatcher per shift
```

```
Baseline: 45 min | Target: ≤ 12 min by day 30</p>
```

```
</blockquote>
```

```
<h3 id="counter-metric">Counter-metric</h3>
```

```
<blockquote>
```

```
<p>% of dispatchers calling support after using the new flow
```

```
If this rises by > 3pp from baseline, we won the primary
```

```
for the wrong reasons.</p>
```

```
</blockquote>
```

```
<h3 id="secondary-metrics">Secondary metrics</h3>
```

```
<ul>
```

```
<li>Reconcile-flow abandonment rate (declining)</li>
```

```
<li>Manual override entries per dispatcher (declining)</li>
```

```
<li>Mobile vs paper adoption ratio (mobile climbing)</li>
```

```
</ul>
```

```
<h3 id="validation-plan">Validation plan</h3>
```

```
<li>Pre-launch: AC verification suite green</li>
```

```
<li>Launch + 7 days: primary directionally correct</li>
```

```
<li>Launch + 30 days: primary at or below 12 min</li>
```

- Launch + 90 days: counter-metric within +3pp band

Activity 1 — §8 Metrics & Validation

Triads • 35 minutes

Pull primary, counter, secondaries from your Tier Sheet. Set targets with defenses. Write the validation plan.

- 10 min: pick metrics from Tier Sheet
- 15 min: set targets, write defenses
- 10 min: validation plan



10 + 15 + 10

Block 2 — §9 Risks, Questions, Assumptions

Honest naming of what could go wrong, what we don't know,
and what we're betting on



The Three Lists

Risks

Things that *could* go wrong. Each gets likelihood, impact, mitigation.

"No risks" is a fail.

Open questions

Things we don't know yet. Each gets an owner and a "needed by" date.

Assumptions

Things we're treating as true. Basis + revisit-if condition.

Watch for: "TBD" as owner. That's the unowned risk that slips. Force a name.



Worked §9 (excerpt)

Risks

```
<table>
<thead>
<tr>
<th>Risk</th>
<th>L</th>
<th>I</th>
<th>Mitigation</th>
</tr>
</thead>
<tbody><tr>
<td>Dispatchers refuse new flow because it's longer</td>
<td>M</td>
<td>H</td>
<td>Optional toggle in v1; observe adoption</td>
</tr>
<tr>
<td>Audit-log volume blows our log budget</td>
<td>L</td>
<td>M</td>
<td>Tier-2 sample for non-error events</td>
</tr>
</tbody></table>
<h3 id="open-questions">Open questions</h3>
<ul>
<li>Will operations VP accept temporary read-only manager view in week 1?
(Owner: Kim; needed by: 2026-05-12)</li>
<li>Can audit store handle 24-month retention at this volume?
(Owner: Pat; needed by: 2026-05-19)</li>
</ul>
<h3 id="assumptions">Assumptions</h3>
```

```
<li>Median dispatcher device is mid-tier Android on 4G LTE
(basis: device-fleet snapshot Q3; revisit if changes)</li>
```

- 30-day adoption signal sufficient to declare success
(basis: prior reconcile rollout; revisit if behavior non-stationary)

Activity 2 — §9 Risks, Questions, Assumptions

Triads • 40 minutes

5 candidate risks → cull to 3–4 with mitigations. 3–5 open questions with owners. 2–4 assumptions with basis.

- 15 min: risks brainstorm + mitigations
- 10 min: open questions + owners
- 10 min: assumptions + basis
- 5 min: assign owners + by-when dates



15 + 10 + 10 + 5

Block 3 — §§10 + 11 Dependencies + Out-of-Scope

Make the network around your PRD explicit

§10 Dependencies

Dependency	Owner	What we need	By when	Status
Auth team — SSO scope	Lin	New scope `reconcile.write`	Sprint 1 Day 1	Confirmed
Data platform — schema	Pat	Schema review for 7 events	Sprint 1 Day 1	Pending
Mobile pipeline — signing	Kim	New signing cert	Pre-launch	Pending
CS — release messaging	Sam	Customer comms	T-7 days	Pending

Owner is a person, not a team. "The auth team" is unowned; "Lin from auth" is owned.

§11 Out-of-Scope Follow-ups

What this PRD acknowledges but won't ship in this iteration:

- **Tablet layout** — same flow, different breakpoint. Ticket: TICKET-1234.
- **Spanish localization** — copy + reading order. Ticket: TICKET-1235.
- **Multi-shop manager view** — cross-shop reconcile. Ticket: TICKET-1236.
- **Offline-first sync** — current ships offline draft + manual reconnect; full offline-first deferred. Ticket: TICKET-1237.

Why this matters: §11 is the negotiation tool for Week 6. "I see what you might also want; here's where it lives in the backlog."

Activity 3 — §§10 + 11

Triads • 40 minutes

Tabulate dependencies (with named owners). List out-of-scope follow-ups (specific, with ticket placeholders).

- 20 min: dependencies (owner / what / when / status)
- 15 min: out-of-scope follow-ups
- 5 min: status sanity check (most should be "Pending"; that's honest)



20 + 15 + 5

Block 4 — Integration Pass + Lock

Read top to bottom; fix incoherence; ship to Friday



The Integration Checklist

Check	Pass criterion
§1 → §2 flow	Reader is motivated to keep reading
§3 goals tied to §8 metrics	Each goal references a metric
§5 sketch ↔ §6 AC	Each happy AC corresponds to a sketch step
§6 AC vs §7 NFRs	NFRs cover what AC don't
§7 NFRs ↔ §8 observability	Every metric has an obs NFR
§4 scope-out → §11 follow-ups	Items left out appear in follow-ups
§9 risks named, owned, mitigated	No "no risks"
No AI-generic prose	Voice is consistent triad voice

✂ AI-Generic Prose Detector

Tells to eliminate:

- "It is important to note that..."
- "By leveraging X, we can unlock Y..."
- "This robust solution will empower stakeholders to..."
- "In today's fast-paced environment..."
- Three adjectives in a row when one would do
- Sentences that sound true but say nothing specific

The test: if a section could appear in any PRD for any product, rewrite it for *your* product.

Activity 4 — Integration Pass + Lock

Triads • 45 minutes • Wrap

Solo read-through, pool issues, fix in priority order, lock. Save as "v0 — for Friday."

- 15 min: solo read-through with margin notes
- 10 min: pool + de-dupe issues
- 15 min: fix coherence first, prose second
- 5 min: lock (Status: In review) + save v0



15 + 10 + 15 + 5 · 15-min wrap-share



Day 4 Wrap

What we shipped

- §§1–11 complete PRD
- Tier-Sheet linked metrics
- Risks owned and mitigated
- Dependencies named
- Integration pass + lock

What opens tomorrow

- **PRD Review day**
- Morning: primary review
- Afternoon: revisions + secondary review
- End of day: signed-off PRD

Bring to Day 5: Your locked v0 PRD. Friday cadence is different — see the morning brief.

Questions & Reflection

Where in your PRD will tomorrow's reviewer push hardest?